

Homework on Flow of Control

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Programming Questions Based on Looping and Selection in Python

1. Student Result Management System

Write a Python program to input marks of 5 subjects for n students.

The program should:

- Calculate total, percentage, and grade
- Use loops for multiple students
- Use selection statements to assign grades
- Display topper details
- Count how many students passed and failed

2. Electricity Bill Generator

Create a Python program to generate electricity bills for customers using the following slab rates:

- First 100 units → ₹5/unit
- Next 200 units → ₹7/unit
- Above 300 units → ₹10/unit

Requirements:

- Use loops for multiple customers
- Apply conditional statements for slab calculation
- Display total bill and category
- Stop when the user enters 0

3. Number Guessing Game

Write a Python program where:

- Computer generates a random number between 1-100
- User keeps guessing until correct
- Program displays:
 - "Too High"
 - "Too Low"
- Count total attempts
- Ask user whether to play again

4. Multiplication Table Generator

Create a program that:

- Accepts a number and range
- Prints multiplication table using loops
- Allows repeated execution
- Uses menu options:
 - Single table
 - Multiple tables
 - Reverse table

5. Employee Salary Processing System

Write a Python program to:

- Input employee name, basic salary, and department
- Calculate:

- HRA
- DA
- Bonus
- Net Salary
- Use conditional statements for bonus rules
- Use loops for multiple employees
- Display highest salary employee

6. Cricket Score Analyzer

Create a Python program to analyze cricket scores.

Requirements:

- Input runs scored in multiple matches
- Calculate:
 - Highest score
 - Lowest score
 - Average
- Count centuries and half-centuries
- Use loops and conditions extensively

7. Voting Eligibility System

Write a Python program that:

- Accepts age of multiple users
- Determines voting eligibility
- Counts:
 - Eligible voters
 - Non-eligible voters
- Stops when age -1 is entered

8. Prime Number Utility

Develop a menu-driven program to:

- Check prime number
 - Print prime numbers in a range
 - Count primes between two numbers
 - Find sum of prime numbers
- Use loops and nested conditions.

9. Bank Interest Calculator

Write a Python program to:

- Calculate simple and compound interest
- Use loops for multiple customers
- Use selection statements for different interest rates:
 - Senior citizen
 - Regular customer
 - Student account
- Display maturity amount

10. Pattern Printing System

Create a menu-driven Python program to print:

- Star pyramid
- Inverted triangle
- Floyd's triangle

- Number pyramid
- Hollow square

Use internet to enquire about the options. Try to use the concepts and solve the problems.

Use nested loops and selection statements.

11. Online Shopping Bill System

Write a program that:

- Accepts product details and quantity
- Calculates total bill
- Applies discounts:
 - Above ₹5000 → 10%
 - Above ₹10000 → 20%
- Adds GST
- Generates final invoice
- Uses loops for multiple products

12. Library Fine Calculation System

Develop a program for library management:

- Input number of late days
- Calculate fine:
 - First 5 days → ₹2/day
 - Next 5 days → ₹5/day
 - Above 10 days → ₹10/day
- Use loops for multiple students
- Display total collection